

Scanner

To **take input from users** we use Scanner class.

Scanner class is a built-in class in the **java.util** package(DTL). Before using the Scanner class you have to import the Scanner class using the import statement as shown below:

To use the **Scanner** class, you need to create an object of it, and then you can use that object to interact with the input data.

```
import java.util.Scanner;  
Scanner sc = new Scanner(System.in) ;//object  
int n = sc.nextInt() ;
```

The nextInt() method parses the token from the input and returns the integer value.

Use methods to read respective data

nextByte(), nextShort(), nextInt(), nextLong(), nextFloat(), nextDouble(), nextBoolean()

Reading String Data

nextLine() - Reads the whole line

next() - Reads the first word

Reading char data – **next().charAt(0)**

Problem with nextLine() method:

If we try to read String after reading in an Integer, Double or Float etc.

Java does not give us a chance to input anything for the name variable.

When the method `input.nextLine()` is called Scanner object will wait for us, to hit enter and the enter key is a character (“\n”).

Example

```
Scanner scanner = new Scanner(System.in);

System.out.print("Enter an integer: ");
int age = scanner.nextInt();

System.out.print("Enter a string: ");
String name = scanner.nextLine();

System.out.println(name + " age is = " + age);
```

Console :

Enter an integer: 69

Enter a string: age is = 69

1. We first prompt the user to enter an integer age using **nextInt()**.
2. After reading the integer, we immediately hit enter and enter is also a character represented by “\n” – 69\n
3. The int value 69 is assigned in age but not the \n still left in the memory or buffer.
4. In next line when we Call **nextLine()** to consume the name it first check in buffer is there any thing as we have \n in buffer it take \n (for nextLine() method \n is the stopping point it will consider we stop giving input and return) and skip the line.

Solution :

```
Scanner scanner = new Scanner(System.in);

System.out.print("Enter an integer: ");
```

```
int age = scanner.nextInt();

// Consume the newline character left in the input buffer
scanner.nextLine();

System.out.print("Enter a string: ");
String name = scanner.nextLine();

System.out.println(name + " age is = " + age);
```

1. After taking an integer input we Call `nextLine()` to consume the name character left in the input buffer. In next line when we Call `nextLine()` to consume the name character left in the input buffer.
2. Then, we prompt the user to enter a string using `nextLine()`.

\ is a special symbol

`\n` (next Line), `\b` (backspace), `\t` (tab), `\"` (double quote), `\'` (single quote), and `\\` (backslash).